

3(a)(i)	<i>the operon has:</i> promoter ; operator ; three structural genes / named three structural genes ; <table border="1" data-bbox="723 486 1552 703"> <tr> <td><i>lacZ</i></td><td>or</td><td>β galactosidase <u>gene</u></td></tr> <tr> <td><i>lacA</i></td><td>or</td><td>lactose / β galactoside, permease <u>gene</u></td></tr> <tr> <td><i>lacY</i></td><td>or</td><td>transacetylase <u>gene</u></td></tr> </table> <i>I regulatory genes / order of named parts</i>	<i>lacZ</i>	or	β galactosidase <u>gene</u>	<i>lacA</i>	or	lactose / β galactoside, permease <u>gene</u>	<i>lacY</i>	or	transacetylase <u>gene</u>	3
<i>lacZ</i>	or	β galactosidase <u>gene</u>									
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<i>lacY</i>	or	transacetylase <u>gene</u>									
3(a)(ii)	<i>any four from:</i> <i>lacI gene</i> 1 is always expressed ; 2 controls (structural) gene expression ; 3 codes for the repressor (protein) ; 4 repressor, binds to the operator / blocks the promoter ; 5 prevents, (structural) gene expression / RNA polymerase binding to promoter ; 6 lactose / allolactose, binds to repressor ; 7 (so) repressor cannot bind to operator / promoter unblocked / gene expression can occur ;	4									

Question	Answer	Marks
3(b)	1 enzymes / proteins, made continuously / all the time ; 2 (because) enzymes / proteins, needed / necessary (for cell) ; 3 end product inhibition / made until product concentrations too high ;	3

Question	Answer	Marks
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